It's the Elicitation, Not the Eyes: Diagnosing VLM Failures in Slide Quality Inspection and Routing a Symbolic–Neural Critic

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Abstract

Vision–language models (VLMs) are now the default critic inside slide- and document-generation agents, yet they routinely pass slides whose text visibly overflows its box. Whether this reflects a failure of *perception* (the model cannot see the defect) or of *calibration* (it sees but will not commit) is rarely asked—and the answer decides whether the remedy is a larger model, a different prompt, or a different tool. We answer it with an attribution protocol built on a *lossless structured-geometry oracle* (element boxes, text, and style, rather than a model-written caption that re-injects perceptual error), and find that VLM "blindness" on rendered documents is largely a protocol artifact with a magnitude-gated floor. Many defects that *look* sub-perceptual under the conventional pointwise+rubric prompt are **format-suppressed**—including fine alignment and color above a small magnitude threshold: the capability is present, and simply *changing the elicitation*—a forced choice, or an atomic per-type query—recovers detection *with the model frozen* (for alignment the paired forced choice saturates to 1.00 once the offset clears $\approx$0.8% of slide width). A **genuine acuity floor** remains only in the sub-threshold tail (a few-pixel offset, a near-isoluminant color shift) and for OCR-bound terminology, where even a side-by-side forced choice stays at chance. Declared geometry routes to a symbolic linter regardless— exact and cheap where the structured IR exists, not because the VLM cannot see the defect—while the format-suppression recovery replicates across the capable models (four of the six VLMs tested, four families) and transfers to third-party human-annotated slides. This diagnosis prescribes the critic: a minimal *symbolic–neural hybrid*—linter for declared geometry plus one C3-prompted VLM for the rest—that covers 8/9 defect classes (balanced accuracy 0.89), matching the pre-registered routed hybrid's 8/9 coverage (0.88) where the best single engine covers 5/9. We sharpen the argument with a *falsifiable* linter-blind render class, G7 (declared box legal, rendered content overflows), and audit *five* published reward scorers (two per category) over it: a symbolic linter (0.00), a document reward (0.48) and an artistic-aesthetic head (0.57) miss it, while *both* general-multimodal rewards (0.79, 0.71 on distinct backbones—a category-level result) and even a zero-shot perceptual-quality probe (0.83) catch it. So G7-detection tracks *perceptual capability, not the training-domain label*, and G7 needs a perceptually-capable engine, whether re-elicited by prompt or carried by a reward head. The diagnosis itself reproduces on *real* CC-licensed slide layouts—a lossless tool-extracted oracle, no annotation—where fine alignment stays near chance in every channel on real cluttered decks (it recovers on clean slides above a magnitude threshold, but real clutter defeats it) while a margin bleed is structure-rescuable. As supporting evidence, an 8B examiner finetuned on injected defects surpasses a zero-shot 30B on *in-distribution* semantic inspection (the edge does not transfer out-of-distribution; Sec. 7), and a perturbation-fidelity audit shows that 45% of injected geometry defects never survive a standard renderer—a hazard any IR-labeled slide dataset inherits. We report the negatives plainly (deck-scope semantics, a sim2real gap, a small downstream effect); each sharpens rather than weakens the thesis.

1 Introduction

An automatic critic that flags layout and content defects is now a load-bearing component of every slide- and document-generation agent, and the default critic is a general-purpose VLM asked to grade a rendered slide. These critics fail in a revealing way: shown a slide whose title visibly spills out of its box, a strong VLM under a standard rubric prompt confidently answers "no defect." Whether this is a failure of *perception* (it cannot see the overflow) or of *calibration* (it

*Independent technical report. All experiments were run on a single 4×RTX 3080 (20 GB) workstation; weights, renders, and raw rollouts are not redistributed but all analysis scripts and reports are released. Correspondence: `surh6@mail2.sysu.edu.cn`.

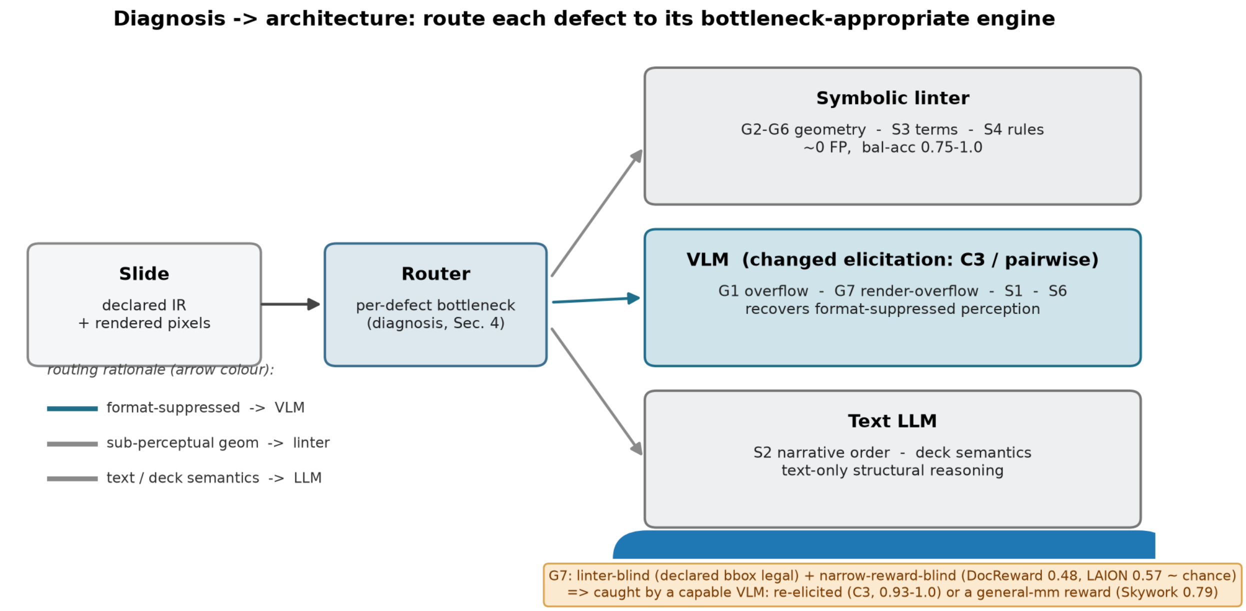

Figure 1: The paper in one picture. A perception/reasoning diagnosis (Sec. 4) shows slide-inspection failures are of two kinds—declared-geometry (exact and cheap for a symbolic linter, plus a sub-threshold acuity tail nothing recovers) and merely format-suppressed (route to a VLM under a changed elicitation). Routing each defect to its bottleneck-appropriate engine yields a hybrid critic (Sec. 5). The render-overflow class G7 is the crux: a symbolic linter is blind to it by construction and a document reward and an artistic-aesthetic head are blind empirically, while *both* general-multimodal rewards (two backbones), a perceptual-quality probe, and the VLM engine under atomic-binary elicitation catch it—so G7-detection tracks perceptual capability, not the domain label, and G7 routes to a perceptually-capable engine.

sees but will not commit) is rarely asked—yet the answer decides whether the remedy is a larger model, a different prompt, or a different tool entirely. We treat that question as the design problem and answer it empirically.

The central observation. A VLM that "cannot see" a slide defect is hiding two failures of opposite character, and separating them is the whole game. Some defects are *genuinely sub-perceptual*: a three-pixel alignment offset or a one-point font mismatch sits below the model's effective acuity, and no amount of prompting, scale, or resolution recovers it. Others are *format-suppressed*: the model can perceive them, but the conventional elicitation—"here is a slide and a rubric; score every defect type in one pointwise pass"—buries the signal, and merely *changing how we ask* (a forced choice between two candidates, or an atomic yes/no per defect type with a required piece of evidence) recovers it with the model frozen. Conflating the two is why the field oscillates between "VLMs are bad at layout" and "VLMs can judge slides"—both hold, for different defects.

Diagnosis with a lossless oracle. We make the split measurable with an attribution protocol built on a *structured-geometry oracle*. Each slide carries an intermediate representation (IR) of element bounding boxes, text, and style; we present the same paired (defective, clean) slides under four input modalities—image-only, structured-oracle, VLM-caption, and image+oracle—crossed with detect/localize/repair tasks. Because the oracle is lossless machine-readable structure rather than a model-written caption, the perception channel it isolates is clean: a model that fails image-only but succeeds with the structured oracle is bottlenecked on perception, not reasoning. This is the methodological difference from concurrent perception/reasoning attribution that uses natural-language perception oracles, which re-inject the very perceptual error they try to factor out.

From diagnosis to architecture. The diagnosis prescribes the critic (Fig. 1): route each defect to the engine that owns its bottleneck. Sub-perceptual geometry goes to a symbolic linter that reads coordinates and rules at near-zero false-positive rate; text and structural semantics go to a language model; and the format-suppressed, perceivable classes go to a VLM, but only under the changed elicitation the diagnosis shows recovers them. To stress-test the design we construct an adversarial class, G7 render-containment overflow, that is invisible to the linter by construction (the declared box is legal; only the rendered pixels overflow) and—we find—invisible to a document reward and an artistic-aesthetic head as well, while *both* general-multimodal rewards (distinct backbones) and a zero-shot perceptual-quality probe see it; G7-detection tracks perceptual capability rather than the training-domain label, isolating exactly the slice of the problem that needs a perceptually-capable engine (prompted VLM or reward head).

Contributions.

- **C1 – A perception/reasoning attribution protocol for generation quality-inspection**, built on a lossless structured-geometry oracle, that separates failures into *sub-perceptual* versus *format-suppressed* (Sec. 4) and reproduces on *real* CC-licensed slide layouts via a tool-extracted lossless oracle—no human or self-annotation (Sec. 7).
- **C2 – "Format suppression, not capability."** Relative and atomic-binary elicitation recover detection that the pointwise+rubric format suppresses; the contrast isolates format from capability, replicates across *the capable models—four of six VLMs tested, four families, three scales*, transfers to real human-annotated slides, and survives a localization check that rules out a lucky "yes" (Sec. 5).
- **C3 – A minimal symbolic–neural hybrid critic and a falsifiable linter-blind class.** The linter+VLM-C3 baseline covers 8/9 classes (0.89), matching the pre-registered routed hybrid (8/9, 0.88) and beating either single engine (linter 5/9, VLM-C3 4/9, VLM-C0 2/9); the render class G7 is caught only by a C3 VLM engine, while a symbolic linter (0.00), a document reward (0.48) and an artistic-aesthetic head (0.57) miss it—yet across *five* audited reward scorers (two per category) *both* general-multimodal rewards (0.79, 0.71, distinct backbones) and a perceptual-quality probe (0.83) catch it, so G7-detection tracks perceptual capability not the domain label, and G7 needs a perceptually-capable engine, prompted or reward-headed (Sec. 5).
- **C4 – A specialist examiner and a perturbation-fidelity audit (supporting).** An 8B examiner finetuned on injected defects surpasses a zero-shot 30B on *in-distribution* semantic inspection (it does not transfer out-of-distribution, Sec. 7) and learns to abstain rather than hallucinate geometry; separately, 45% of injected geometry defects never render under a standard template renderer—a hazard any IR-labeled slide dataset inherits (Secs. 6, 5).

This is a technical report, not a leaderboard entry. We have deliberately included the experiments that *cut against* the thesis—a near-zero downstream effect under a strong generator, a degenerate deck-scope head, and a synthetic-to-real transfer gap on bare pixels—because each one tells us something about where the perception/calibration boundary actually lies.

2 Related work

The VLM perception bottleneck. A growing line of work argues that VLM failures on visually grounded tasks are dominated by perception rather than reasoning—decomposing the bottleneck for chart understanding [1], or showing that the needed visual information is present in the representation but unused [2]; MMVP [3] is the canonical demonstration that VLMs stumble on minimally-different image pairs humans solve trivially—the sub-perceptual acuity floor our title ("not the eyes") deliberately echoes. We share the perception-first stance but ask it of a different task family—*quality inspection of generated artifacts*—where the perception channel can be isolated exactly with structured geometry rather than approximated by captions, and where the answer feeds a critic architecture rather than a benchmark.

Perception/reasoning attribution. Concurrent work separates perception from reasoning in abstract visual-reasoning benchmarks via natural-language perception oracles [4]. We differ in three ways: (i) our domain is inspection of generated documents, which admits a controllable defect taxonomy with *exact injected labels*; (ii) our perception oracle is a *lossless structured representation*, not a lossy caption, yielding a cleaner attribution boundary; and (iii) we close the loop from diagnosis to a deployed critic. Layout-error-detection work [5] pioneers distribution-grounded structural-error injection but for *document-layout-analysis predictions*, and explicitly notes that existing metrics cannot isolate whether failures stem from recognition or reasoning—precisely the gap our attribution protocol fills.

Relative vs. absolute judgement. Our finding that relative elicitation beats absolute scoring echoes work showing VLM judges can rank but cannot reliably score [6]; that result concerns a judge's *output mode*, orthogonal to our *source of failure*, and we adopt it to motivate the pairwise and atomic-binary heads of our critic. More broadly, comparative/relative elicitation is known to recover discriminative signal that pointwise scoring suppresses in LLM/VLM judges [7], even though the protocol choice is not bias-free and can itself distort preferences [8]; our contribution is not

Table 1: Defect taxonomy. The geometry group is detected symbolically; the semantic group is the examiner's domain; G7 is constructed to be linter-blind. "Channel" is the engine the diagnosis assigns (Sec. 4–5).

Group	Class	Description	Channel
Geometry	G1 text overflow	text exceeds its box	VLM (pairwise) + linter
	G2 element overlap	boxes intersect	linter
	G3 alignment offset	small mis-alignment	linter
	G4 font-size inconsistency	off-grid font size	linter
	G5 brand-color violation	off-palette color (ΔE)	linter
	G6 margin violation	element past safe margin	linter
Semantic	S1 title/body mismatch	body contradicts title	VLM
	S2 narrative-order break	deck sections out of order	LLM
	S3 terminology inconsistency	term drift across deck	term linter
	S4 density violation	slide over/under-packed	LLM
	S5 missing logic section	dropped argument step	LLM
	S6 image/text contradiction	figure contradicts caption	VLM
Render	G7 containment overflow	legal box, rendered overflow	VLM (atomic-binary)

that effect but its *defect-class boundary*—it rescues G1/S6 yet not G3/S3—which is what turns the elicitation effect into a perception/format attribution signal.

Slide generation, evaluation, and reward. Slide and poster generation with design feedback is an active area; aesthetic-reward systems [9, 10] report that iterative VLM refinement helps gross layout damage but fails on fine-grained aesthetics and is prone to reward hacking, and slide-evaluation studies [11] call for calibrated critic-in-the-loop selection. These motivate, rather than compete with, a routed critic: they observe the VLM's perceptual ceiling and route around it with rules, whereas we first *diagnose* that ceiling and then show part of it is recoverable by elicitation. We use the human-annotated SlideAudit set [12] as our real-data probe and audit *five* published reward scorers (two per audited category)—the document reward DocReward [13]; two general-multimodal rewards on distinct backbones, the value-head Skywork-VL-Reward [14] and the CLIP-contrastive PickScore [16]; and two aesthetic/perceptual scorers, a LAION aesthetic CLIP predictor [15] and the zero-shot CLIP-IQA quality probe [17]—as neural-critic baselines.

Critic / reward-model training. Training specialist critics from paired data is established for natural images and code; our examiner adapts the recipe to the rendered-document domain, where defect injection yields exact labels at zero annotation cost. We are careful *not* to claim injection itself as novel—it is a known recipe—only its use as the semantic engine of a routed critic and the perception/reasoning attribution it enables.

3 Problem setup and defect taxonomy

Representation and taxonomy. Every slide is an IR of typed elements with bounding boxes, text, and style; defects are injected programmatically into the IR and then rendered, giving exact labels and matched (defective, clean) pairs. The taxonomy (Table 1) has a *geometry/perception* group G1–G6, a *semantic/reasoning* group S1–S6, and one constructed render class G7; the taxonomy is aligned to the expert-derived SlideAudit taxonomy [12] via a bidirectional map (results are reported in its canonical classes), with G7 refining its "content overflow/cut-off" category. We add G7 because it is the sharpest test of the linter's blind spot: its declared box is legal (inside the safe margin, no box overlap) but the rendered content overflows its container, so the defect exists *only* in pixels.

Attribution modalities and metric. We present each paired sample under four modalities: **A** image-only; **B** the lossless structured oracle (with ground-truth markers stripped by a single de-leaking view, regression-tested); **B′** a VLM-written caption held fixed across models; and **C** image+oracle—crossed with detect/localize/repair. The attribution logic is: A-fail $\wedge$ B-succeed $\Rightarrow$ *perception* bottleneck; A-fail $\wedge$ B-fail $\Rightarrow$ *reasoning* bottleneck; A-succeed $\wedge$ repair-fail $\Rightarrow$ *execution* bottleneck. Throughout, the headline metric is *balanced accuracy on paired clean controls*, never recall alone—a consistency defect scored by recall reads a false "100% detection" when the model simply always flags.

Two linters. A geometry linter detects G1–G6 from coordinates and rules (brand color via CIELAB ΔE_{2000}, not RGB distance), at zero false positives on clean slides. A terminology linter handles S3 from an occurrence table plus near-duplicate clustering, image-free. Both are the "detectors of record" the diagnosis routes to.

4 Diagnosis: two kinds of blindness

Pointwise image-only inspection floors fine geometry. Under pointwise image-only inspection, G2–G6 sit at chance for 4B and 8B models, and a 30B model breaks only the coarsest class (G1 overflow). This pointwise floor is invariant to the things one would tune first: swapping the vision encoder across *five* families (SigLIP2, an LLM-based encoder, InternViT, a native-resolution NaViT, and MoonViT) leaves every model at the random line, differing only in abstain-vs-over-flag bias, and raising input resolution from 1536 to 2048 changes nothing. The lever is the elicitation and the defect magnitude, not the encoder or the pixel budget (the next paragraph and Fig. 3 show targeted/relative elicitation lifts supra-threshold alignment well above chance)—so the floor is a property of the *pointwise* protocol, not a hard capability limit. Either way the right detector for declared geometry is the symbolic linter: it reads the coordinates directly at balanced accuracy 0.75–1.0 with zero false positives—cheaper and exact where the IR is available, not because the VLM cannot see the defect.

But some "perceptual" failures are elicitation artifacts, not blind spots. The same pointwise image-only setting hides a second, opposite phenomenon. For G1 text overflow and S6 image/text contradiction, pointwise balanced accuracy is 0.50—indistinguishable from the genuinely sub-perceptual classes—yet a two-alternative forced choice between the defective and clean slide jumps to **1.00** (Fig. 2): the defect is perceivable, and the absolute pointwise call simply fails to elicit it. The same is true of G3 alignment, but only *above a magnitude threshold*: when we sweep the injected offset as an internal contrast (one element shifted out of its aligned sibling column, decidable from the slide alone) across 2–64 px (0.1–3.3% of slide width), the forced choice saturates to **1.00 by 16 px** (0.83%) and single-slide atomic elicitation climbs to a ≈ 0.90 plateau by 48–64 px, while the *sub-threshold tail* ($\leq$ 8 px) stays at chance even side by side (Fig. 3). So fine alignment is not a class-level capability floor: it is format-suppressed and recoverable wherever the offset clears acuity, with a genuinely sub-perceptual residue only in the small-offset tail. (*How* the elicitation helps differs by class—a named target on a single slide vs. a paired clean reference—and we decompose it in Sec. 5.1 (Fig. 6), finding pure format suppression for the constructed render class G7 and a milder availability-of-reference effect for G1/S6.) The connective thread is therefore a *magnitude-gated* recoverable-versus-sub-perceptual boundary rather than a clean geometric/semantic split—and S3 terminology (below) remains the one defect no elicitation rescues.

One honest counterexample: terminology. Not every consistency defect is format-suppressed. S3 terminology inconsistency is *not* rescued by forced choice (0.625, heavy position bias): reading the same term across deck pages and spotting a subtle variant is below threshold even side by side. Tellingly, the structured-oracle channel—which feeds the model the full deck text with both variants present—also fails (0.56), so the bottleneck is OCR-from-pixels plus the yes/no framing, not reasoning. S3 therefore leaves the VLM entirely for a symbolic terminology linter, which reaches balanced accuracy 1.000 against the VLM's 0.69. A negative result that *re-routes* a class is exactly the kind the diagnosis is for.

The falsifiable counterpart: a genuine geometric blind spot. The re-operationalisation matters only if it can also *fail* to rescue—otherwise "change the elicitation" would be an unfalsifiable escape hatch. It fails cleanly on one class. We re-pose G6 margin as a *page offset*: the whole content block is shifted toward one edge—every element together, so internal alignment is preserved and the defect cannot be misalignment—leaving one margin crowded against (or running off) the edge and the opposite side empty, decidable from the slide alone.[1] Across all six VLMs and both pointwise framings (absolute "past the safe margin" and relative "shifted to one side") the page offset stays at chance (roster-mean C3 0.50), and the discriminative 2-AFC—the very test that separated *data artifacts* (S6 image/text contradiction, which recovers to perfect once the figure is actually rendered) from genuine blindness—leaves G6 on the floor for *every* model (Fig. 4): none can tell a page-offset slide from a balanced one until the content is clipped by the page boundary. So unlike G3/G5 (format-suppressed, recoverable) and S6 (a rendering artifact, recoverable), page offset is a *genuine*

[1] The original S6 render omitted the figure element, so the cross-modal contradiction was literally unseeable (recall 0); on the valid-figure corpus the S6 2-AFC reproduces at 0.75–1.00 across all six VLMs, confirming it is a render artifact, not blindness.

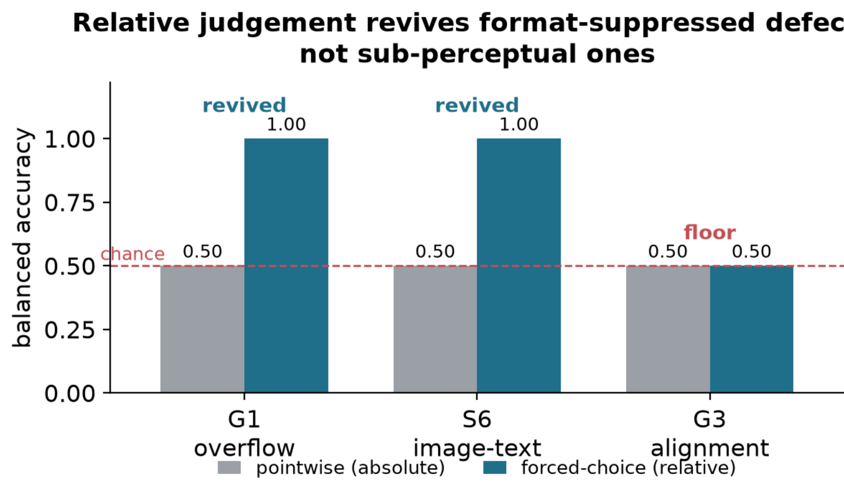

Figure 2: Relative judgement recovers absolute-pointwise blind spots. For G1 overflow and S6 image/text contradiction, pointwise (absolute) inspection is at chance but a forced choice recovers perfect detection (*elicitation-recoverable*). G3 alignment behaves the same way above a magnitude threshold (forced choice 1.00 once the offset clears $\approx 0.8\%$ of slide width; Fig. 3); only its sub-threshold tail and S3 terminology stay at chance side by side. Same model, same images; only the elicitation changes. What *drives* the recovery (naming vs. a paired reference) is decomposed in Fig. 6.

perceptual blind spot—VLMs do not represent global content positioning—which is exactly why declared geometry routes to the linter.

A methodological aside that becomes load-bearing. Enterprise "snap-to-master" templates re-fit element boxes to a grid before rendering. Measuring this, we find a template silently *absorbs* 45% of injected geometry defects: overlap, alignment, and margin violations become pixel-identical to their clean twins after snapping (Sec. 5). This collapses the inspectable spectrum toward two poles—pure-perception defects that templates cannot constrain (overflow, driven by content-determined text length) and pure-semantic defects that templates do not touch—and it is a trap for any dataset that derives labels from the IR but evaluates on the rendered image. We use it below to explain a reward-model failure and to justify rendering defectives faithfully.

5 A routed symbolic–neural critic

The diagnosis says a single model is the wrong unit: route each defect to the engine that owns its bottleneck. We build that critic, validate the elicitation recovery it relies on across many models, and stress it with the constructed render class G7.

5.1 Elicitation: format suppression is not capability

Four ways to ask. We compare four elicitations holding the model and image fixed: **C0**, the conventional pointwise+rubric pass over the whole taxonomy in one call; **C1**, free-form describe-then-classify; **C2**, a geometry-normalized synthetic-twin pairwise (re-render snapped-to-grid, both orders); and **C3**, an atomic per-type yes/no with forced localization. The scientific contrast is **C3 vs. C0**: same model, taxonomy, and image, differing only in "ask everything at once" versus "one atomic binary with forced evidence." If C3 $\gg$ C0, the pointwise abstention was a format/cognitive-load artifact, not a capability gap.

The recovery replicates across families and scales. On the constructed render class G7, the atomic-binary C3 recovers detection that C0 suppresses, and it does so across four model families and three scales (Fig. 5): Qwen3.5-9B $0.50 \to 0.93$, Qwen3.5-27B $0.93 \to 1.00$, Qwen3.6-27B $0.52 \to 1.00$, and Google Gemma4-31B $0.75 \to 1.00$, all at precision 0.88–1.00. As a *cross-model* effect—the claim we actually make—the recovery is overwhelming: a stratified (Cochran) McNemar over the four models' matched pairs counts **149** G7 items that C3 fixes against **zero** reversals

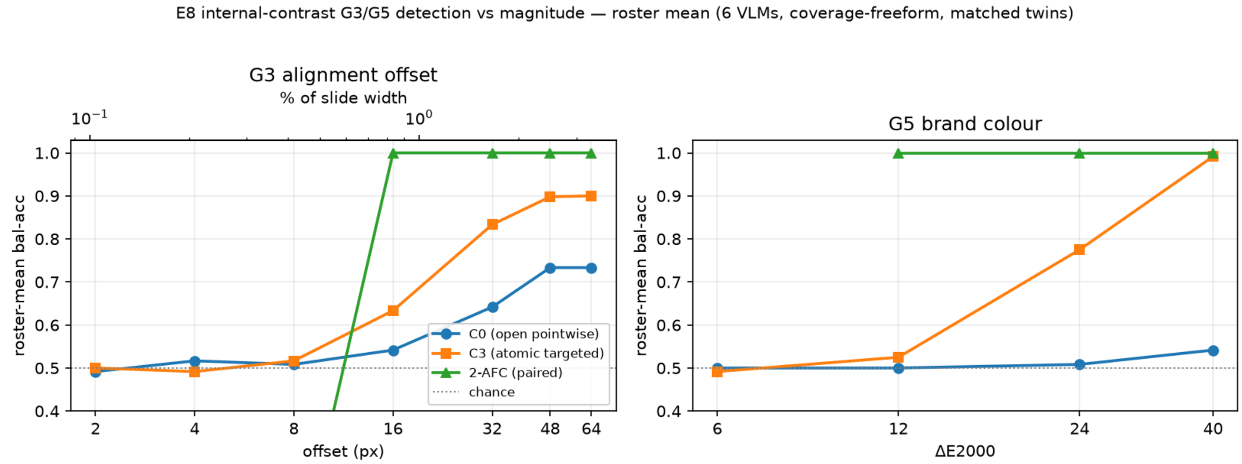

Figure 3: Fine geometry is magnitude-gated, not a flat capability floor. Roster-mean balanced accuracy (6 VLMs, matched-twin internal-contrast set) versus injected magnitude for G3 alignment (left; px and % of slide width) and G5 brand color (right; ΔE_{2000}). Open pointwise (C0) lags; atomic targeted elicitation (C3) recovers to a ≈ 0.90 plateau by 48–64 px / 0.99 by ΔE 40; the paired 2-AFC saturates to 1.00 by 16 px (0.83%) / ΔE 12. The sub-threshold tail ($\leq$ 8 px, $\leq 6\,\Delta E$) stays at chance under every protocol—the genuinely sub-perceptual residue. Single-slide C3 plateaus below 1.0, a residual ceiling on *absolute* spatial judgement that contrastive elicitation does not share.

(χ^2_1=149, $p < 10^{-30}$).[2] That the *same* effect appears in models from different vendors with different pretraining is the core model-agnostic evidence: C0's fixed taxonomy cannot *name* an off-taxonomy render class, so it abstains; the atomic per-type binary restores it.

It is real perception, not a lucky "yes." Paired-clean precision near 1.00 already rules out an always-flag model, but to confirm a correct "yes" points to the right place we check C3's forced evidence against the synthetic G7 ground truth. On the four capable models' G7 true positives, the named region and the named overflowing *element* are correct ≥98%; models read back the specific spilling content ("the list items starting from 'Cost attribution' "), which a bluffing model cannot do. The recovery is perception, not a calibration trick on the label.

No free lunch: elicitation is conditional. The optimal elicitation is jointly determined by model capability and defect class. C2 synthetic-twin pairwise—not C3—is what rescues *declared* geometry G1 on capable models (Qwen3.5-27B 0.69 → 0.93, Qwen3.6-27B 0.53 → 0.78, precision 1.00), because snapping absorbs the declared overflow and the pairwise contrast exposes it. And C1 free-form, which rescues weak abstaining models, *collapses* on strong models: the capable, verbose models free-associate nits and flag a defect on nearly every clean slide (precision ≈0.50). There is no universal elicitation—which is precisely why the critic routes per defect and per model rather than adopting one prompt.

Naming vs. a paired reference: a defect-dependent answer. A skeptic asks whether the chance→1.00 recovery is genuine *elicitation* (the model perceives the defect and the format suppressed it) or merely an *easier task* (the forced choice hands over a clean reference and a named target the pointwise call withholds). We factor it, holding model and image fixed, into a *naming* term (C0_named: a single slide, the candidate defect named, no reference, no forced-evidence gate) and a *pairing* term (the true 2-AFC against the clean twin), with a clean-vs-clean *guess-floor* control (Fig. 6). The answer is *defect-dependent* and consistent across all four capable models. For G7—the off-taxonomy render class C0 cannot *name*—**naming alone** recovers it (Qwen3.5-9B 0.50 → 0.93, Qwen3.6-27B 0.51 → 1.00, Qwen3.5-27B 0.79 → 1.00, Gemma4-31B 0.69 → 0.88; paired-clean McNemar Holm-corrected $p \leq 0.002$), and the paired reference adds nothing on top: this is *format suppression*, the model saw G7 all along. For G1/S6 the opposite holds: naming

[2] Per model the gain is individually significant for the three low-C0 models ($p \leq 2\times10^{-9}$); on the already-strong Qwen3.5-27B (C0 0.93) only 9 items can move and all 9 go to C3 (0 reversals, exact p=0.004, individually n.s. after Holm over the full 61-test family)—a ceiling effect, not a fragile one. Holm (FWER) over all 61 reported tests keeps the pooled G7 recovery, the three low-C0 G7 cells, both examiner contrasts (p_{Holm}=1.9×10^{-5}, 4.7×10^{-4}) and *both* general-multimodal rewards' G7 detection (Skywork p_{Holm}=2×10^{-6}, PickScore 2.6×10^{-3}); full corrected table + the stratified test in the released multiplicity report.

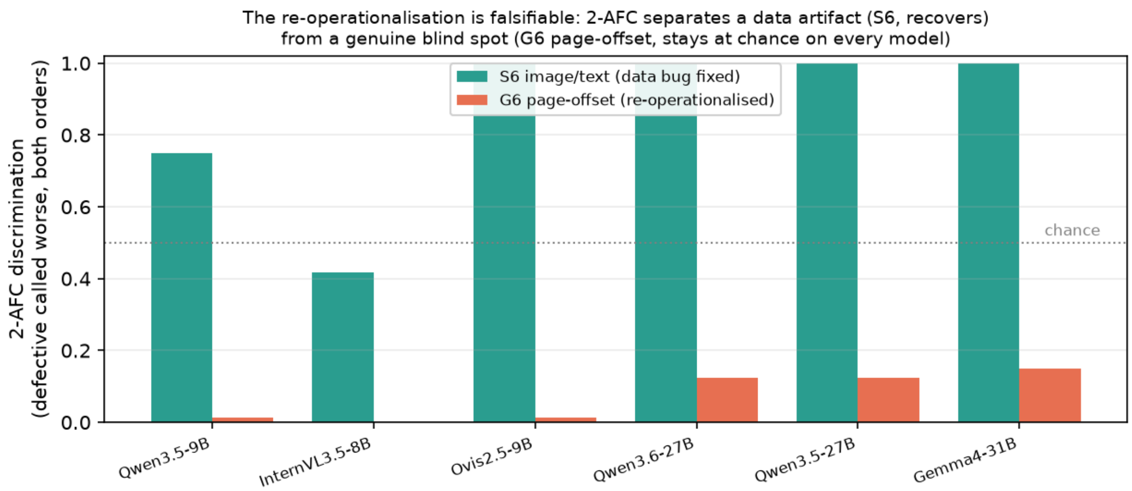

Figure 4: The re-operationalisation is falsifiable. Per-model 2-AFC discrimination (fraction of paired slides where the defective is consistently called worse in both presentation orders; ties count as not-discriminated). Single-image pointwise puts both classes at chance, but the forced choice *separates* them: the data artifact S6 (image/text contradiction, on the fixed valid-figure corpus) recovers on every model, while the re-operationalised G6 page offset stays at the floor on every model—a genuine blind spot, not an elicitation or data artifact.

alone does *not* help (C0_named ≈0.50) and the recovery needs the paired clean reference (2-AFC →0.97–1.00)—an *availability-of-reference* effect, properly scoped as such. Crucially, the guess-floor control rules out a forced-pick artifact for both: on two *clean* slides the capable models correctly answer "tie" 92–100% of the time and fabricate a consistent winner on ≤2% of pairs, so the 2-AFC ceiling is real signal, not always-pick-one. The paper's mechanism claim—*format suppression, not the eyes*—is thus established for the constructed G7 (and the C3 atomic recovery it mirrors); the declared G1/S6 forced-choice recovery is the milder, honestly-labelled availability-of-reference effect.

It transfers to real slides. On the human-annotated SlideAudit set, even with no structure available (so no linter), C3 beats C0 pointwise for every class—brand color 0.52 → 0.81, alignment 0.56 → 0.72—confirming the format effect is not a synthetic artifact. Absolute geometry on bare real pixels stays modest (0.55–0.74 outside blatant overlap): real fine geometry still needs the linter, which needs the element structure bare images lack. The hybrid's full power requires native structure; on image-only real data it runs in a degraded VLM-only mode—a ceiling set by missing structure, not missing method.

5.2 The minimal hybrid critic covers the most

Static routing, complementary engines. We evaluate five critics by *named attribution*: the correct defect must be emitted on the defective slide and not on its clean twin. The E3 control is decisive: the minimal hybrid—linter for declared geometry plus one C3-prompted VLM elsewhere—covers **8/9** classes at **0.89**, matching the pre-registered routed hybrid (**8/9**, **0.88**) and beating either single engine (linter: **5/9** at 0.69; VLM-C3: **4/9** at 0.72; VLM-C0: **2/9** at 0.59; Fig. 7, Table 2). The nine classes are the image-level paired-clean classes; G4/S3 are linter-owned and S2/S5 are deck-scope (Sec. 6). Thus the falsification branch strengthens the message: per-class routed prompt/LLM tuning is not load-bearing here; *linter* ⊕ *C3-VLM* is the cleaner result.

A data-driven routing correction, reported as such. We initially routed S1 title/body mismatch to the text LLM, but the text-only probe over-flags (0.25, precision 0.09) while the image-bearing VLM names it reliably (0.94, precision 0.90): title/body mismatch needs the *rendered layout*, not just the text. We re-route S1 to the VLM—and specifically to its *open* pointwise prompt (C0, 0.94) rather than the atomic C3 (0.83): recall is already saturated on S1 (both elicitations catch all 18 defectives), so the forced yes/no of C3 only adds false positives, dropping specificity 0.89 → 0.67. Atomic elicitation is therefore *not* a free lunch—it recovers recall-limited classes (G7, S4) but inflates false positives on subjective ones—which is itself the reason the critic routes the *elicitation* per class instead of forcing C3 everywhere.

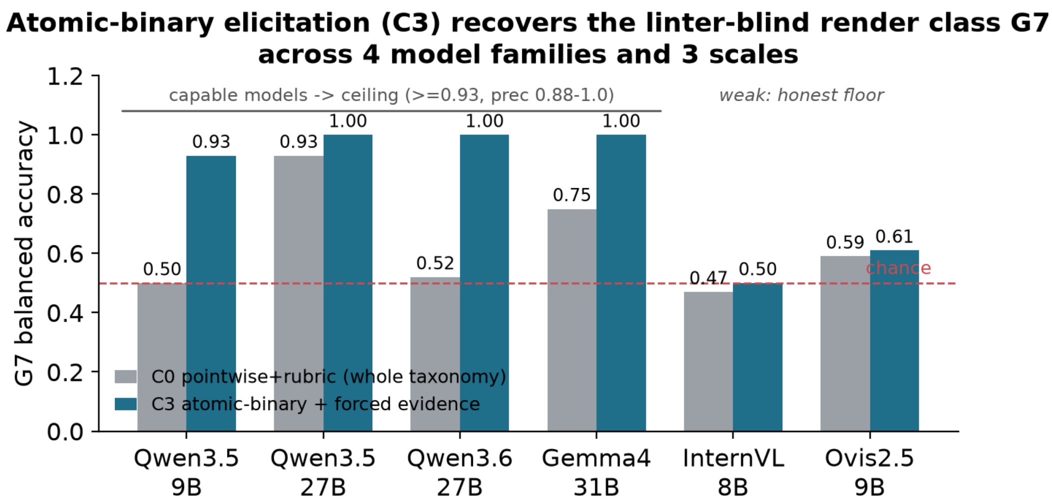

Figure 5: Atomic-binary elicitation (C3) recovers the linter-blind render class G7 across families. Capable models reach the ceiling (≥0.93, precision 0.88–1.0); the weaker InternVL/Ovis stay near their perceptual floor—an honest, capability-dependent boundary, not a universal fix.

Table 2: Per-class named-attribution balanced accuracy with 95% Wilson intervals. E3 adds C3-everywhere and linter+VLM-C3: the minimal hybrid matches routed coverage and slightly exceeds its mean. n = paired positives with matched clean controls; precision for every cell is in the released reports.

Class	routed to	linter	VLM-C0	VLM-C3	linter+C3	routed	n
G1 overflow	linter	1.00 [.91,1]	0.50 [.46,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G2 overlap	linter	0.80 [.68,.87]	0.71 [.60,.79]	**0.86 [.74,.92]**	0.80 [.68,.87]	0.80 [.68,.87]	40
G3 alignment	linter	0.93 [.85,.96]	0.62 [.55,.68]	0.71 [.63,.77]	**0.93 [.85,.96]**	**0.93 [.85,.96]**	70
G5 color	linter	1.00 [.91,1]	0.50 [.46,.54]	0.68 [.57,.75]	**1.00 [.91,1]**	**1.00 [.91,1]**	40
G6 margin	linter	1.00 [.90,1]	0.50 [.45,.55]	0.50 [.45,.55]	**1.00 [.90,1]**	**1.00 [.90,1]**	36
G7 render-overflow	VLM (C3)	0.00 [0,.54]	0.50 [.46,.54]	**1.00 [.91,1]**	**1.00 [.91,1]**	**1.00 [.91,1]**	40
S1 title/body	VLM (C0)	0.50 [.41,.59]	**0.94 [.75,.98]**	0.83 [.63,.92]	0.83 [.63,.92]	**0.94 [.75,.98]**	18
S4 density	LLM	0.50 [.45,.55]	0.51 [.44,.59]	**0.93 [.81,.97]**	**0.93 [.81,.97]**	0.72 [.60,.80]	36
S6 image/text	VLM	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	0.50 [.41,.59]	18
mean / classes covered		0.69 / 5/9	0.59 / 2/9	0.72 / 4/9	**0.89 / 8/9**	0.88 / 8/9	

We also do not hide the one image-level class the deployable hybrid fails to cover: S6 image/text contradiction stays at 0.50 single-image for every engine—but the 2-AFC diagnosis (Fig. 4) shows the model *does* perceive it (it over-asserts pointwise rather than missing it; recall 1.0, specificity ~ 0), so the gap is availability-of-reference, not blindness—in pointed contrast to the genuine G6 page-offset blind spot.

5.3 G7: a falsifiable class that the linter and specialised rewards miss

The construction. G7 is defined to make the linter's blind spot falsifiable: a declared box that is legal (in-margin, non-overlapping) but whose rendered content overflows its container. The IR carries only legal short text; the overflow lives in the renderer, not the IR, so the linter and any structure-only model are blind by construction. A self-check confirms the geometry linter returns zero findings on 90/90 G7 defectives.

What kind of reward sees G7? Perceptual capability, not the training-domain label. The natural rebuttal is "just train a neural reward model." We test it directly by auditing **five** published reward scorers—now *two per audited category*, so the answer is category- not instance-level—on the same paired slides: does each score the clean slide above

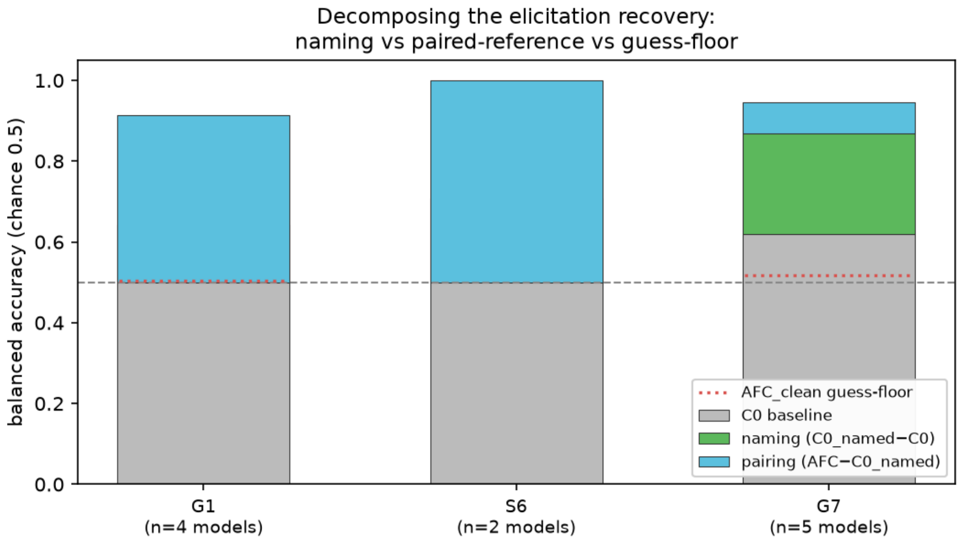

Figure 6: What drives the recovery: naming vs. a paired reference. Holding model and image fixed on the freeform set, the chance→ceiling jump factors into a *naming* component (C0_named−C0: one slide, the defect named, *no* reference) and a *pairing* component (AFC−C0_named: a clean twin supplied). For the constructed render class G7 naming alone carries it (green); for declared geometry G1 and semantic S6 the paired reference does (blue). The AFC_clean guess-floor (dotted) is near chance, so neither is a forced-pick artifact. Mean over the models that recover (AFC $\geq$ 0.7).

its defective twin? The scorers are a *document* reward (**DocReward** [13], Qwen2.5-VL-3B with a Bradley–Terry head over 117K paired documents); two *general*-multimodal rewards on *distinct* backbones (**Skywork-VL-Reward-7B** [14], a Qwen2.5-VL-7B value head, and **PickScore** [16], a CLIP-H/14 contrastive reward fine-tuned on 500K human choices); and two *aesthetic*/perceptual scorers (**LAION-Aesthetic** [15], a linear artistic-appeal head on CLIP-L/14, and **CLIP-IQA** [17], a zero-shot "Good/Bad photo" quality probe on the same CLIP). Each cleanly penalizes several in-taxonomy defects it can see (Table 3, Fig. 8), so each is a working scorer, not a domain mismatch. On G7 the picture splits—but *not* along the domain label. G7 is detected (95% CI above chance, n=90) by *both* general-multimodal rewards (Skywork-VL 0.79, $[0.69, 0.86]$; PickScore 0.71, $[0.61, 0.80]$)—a *category-level* result across two unrelated backbones—and, tellingly, by the zero-shot perceptual-quality probe (CLIP-IQA 0.83, $[0.74, 0.90]$); it is missed (CI spans 0.5) by the symbolic linter (0.00), the document reward (DocReward 0.48, $[0.38, 0.58]$) and the artistic-aesthetic head (LAION 0.57, $[0.46, 0.66]$). So the dividing line is not "general vs. narrow domain" but *perceptual capability*: a scorer that reads generic rendered quality—a capable general-multimodal reward, or even a plain CLIP quality read—catches the visible overflow, whereas a head tuned to a *specialised* objective (document structure, artistic appeal) is blind to it even when, as for CLIP-IQA vs. LAION, the two share a backbone. This is the reward-side counterpart of the C0→C3 recovery: the perception is present; a specialised read-out suppresses it. The hybrid thesis is unchanged and now category-level: G7 needs a perceptually-capable engine, and that engine works whether it is *prompted* (the C3 recovery, 0.93–1.00) or carries a *reward head* (two general-mm rewards, 0.71–0.79)—a reviewer who tests either will find detection, and we report the full split, including the aesthetic scorers that disagree. (Naming containment in Skywork's prompt lifts G7 to 0.87, reinforcement rather than rescue. On G5—re-operationalised as an internal chromatic clash, one element hue-swapped against its aligned sibling column and so visible from the slide alone (Sec. 4)—*most* scorers now react (0.65–0.78; only Skywork is marginal at 0.57), exactly as the perceptual-capability reading predicts for a visible defect: a chromatic inconsistency is *not* the linter-only blind spot G7 is. The linter still owns G5 at 1.00, but its advantage there is exactness, zero false positives, and knowledge of the *declared* palette—not that the clash is imperceptible.)

Perturbation fidelity: 45% of injected geometry never renders. The reward audit ties to the template result of Sec. 4. Rendering every injected defective two ways—faithfully and snapped-to-master—we find 45% of injected geometry defects (100% of overlap, alignment, margin; 20% of overflow) become pixel-identical to their clean twin after snapping. Feeding the *snapped* pairs to *all five* reward scorers gives preference accuracy 0.00 at a reward gap of exactly 0.000: the two images are pixel-identical, so no critic—document, general-multimodal, or aesthetic—can

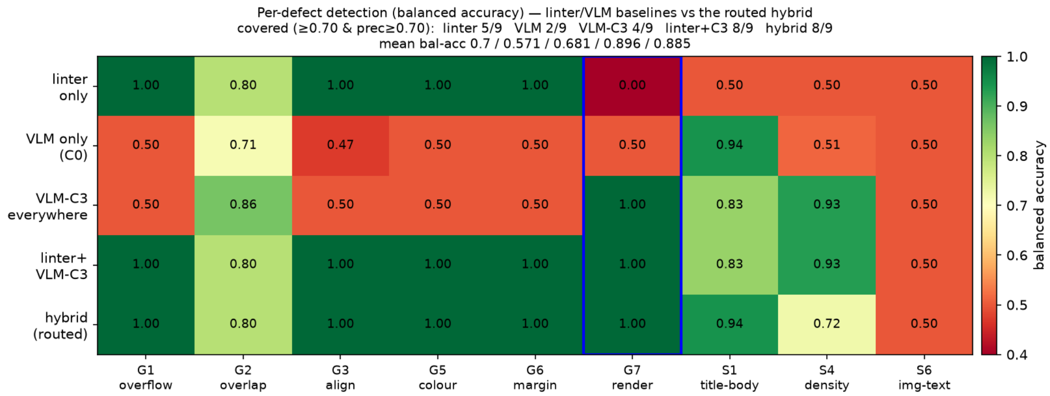

Figure 7: Coverage by engine. Per-defect balanced accuracy for the linter, VLM-C0, VLM-C3, linter+VLM-C3, and the routed hybrid. Minimal linter⊕C3 matches routed coverage; G7 (boxed) is the linter-blind cell rescued by C3.

Table 3: Multi-reward audit: paired preference accuracy $P(r_{\text{clean}} > r_{\text{def}})$ per class (chance = 0.5; G7 with 95% CI, n=90). **Five** published scorers, now *two per audited category* (general-multimodal: Skywork-VL + PickScore on *distinct* backbones; aesthetic: LAION + CLIP-IQA), so the G7 read is category- not instance-level. G7 (boxed in Fig. 8) is detected (CI above 0.5) by *both* general-mm rewards and by the zero-shot perceptual-quality probe (CLIP-IQA), and missed (CI spans 0.5) by the document reward and the artistic-aesthetic head—so detection tracks perceptual capability, not the training-domain label. ✓/× in the G7 column marks reliable detection. G5 is the internal-chromatic re-operationalisation of Sec. 4 (one element hue-swapped vs. its sibling column, n=80), a *visible* defect most scorers catch—not the linter-only blind spot G7 is.

Reward scorer	category	G1	G2	G5	G7	S4	S6
DocReward-3B	document	0.93	1.00	0.72	× **0.48** [.38, .58]	1.00	1.00
Skywork-VL-7B	general-mm	0.94	0.72	0.57	✓ **0.79** [.69, .86]	0.92	1.00
PickScore-v1	general-mm	0.74	0.59	0.70	✓ **0.71** [.61, .80]	1.00	0.72
LAION-Aesth.	aesthetic	0.37	0.91	0.78	× **0.57** [.46, .66]	0.75	0.61
CLIP-IQA	aesthetic	0.44	0.50	0.65	✓ **0.83** [.74, .90]	0.50	0.72
n (paired)		54	54	80	90	36	18

react—a dataset-integrity hazard (identical inputs force identical scores), not a reward-model deficiency. This part *is* model-agnostic by construction, and any reward model trained or evaluated on snap-rendered images with IR-derived labels inherits these zero-signal pairs. Perturbation fidelity must be checked at the pixel level, not assumed from the IR—a prerequisite the field largely skips, and the reason our elicitation experiments render defectives faithfully.

6 The semantic engine: a specialist examiner

The hybrid's semantic engine can be a zero-shot model, but a small specialist does better on its own ground. We finetune Qwen3-VL-8B with QLoRA on programmatically injected defects (2,142 routed SFT records: semantic pointwise, geometry restate-from-structure with image-only abstention, and G1/S6 pairwise), and evaluate on held-out splits.

An 8B examiner surpasses a zero-shot 30B on *in-distribution* semantics. On in-domain held-out page-semantic inspection, the finetuned 8B reaches balanced accuracy 1.000 on both S1 title/body and S4 density, beating zero-shot 8B (0.78 / 0.53) and zero-shot 30B (0.93 / 0.77) (Table 4). The largest gain is S4 density recall, 0.0→0.97, where the finetuned 8B (1.00) beats the zero-shot 30B (0.65) at two-proportion z, $p = 4.5 \times 10^{-7}$: an 8B examiner surpasses a 4× larger zero-shot model on the examiner's core job. The advantage is not a prompt-format artifact—re-running the zero-shot baselines at the finetuned model's own trained prompt format collapses them to 0.50, so the format asymmetry favors the baselines, and the finetuned model still wins at each model's best format. On held-out *severities* the semantic edge holds (0.89 vs. 0.50–0.59).

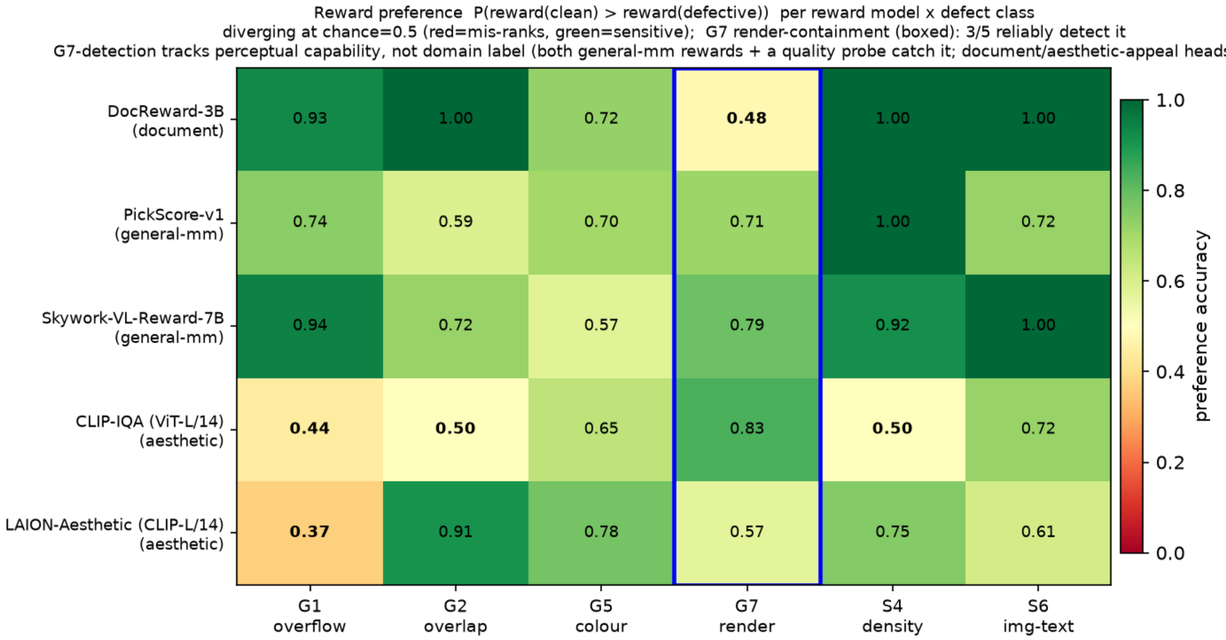

Figure 8: G7-detection tracks perceptual capability, not the training-domain label. Preference accuracy per reward model × defect class (green = sensitive, red = mis-ranks; chance = 0.5), five scorers, two per audited category. Every scorer is sensitive to several in-taxonomy classes, but for the boxed G7 render class detection splits by *capability*: both general-multimodal rewards (Skywork-VL 0.79, PickScore 0.71, distinct backbones) and the zero-shot perceptual-quality probe (CLIP-IQA 0.83) catch it, while the document reward (DocReward 0.48) and the artistic-aesthetic head (LAION 0.57) sit at chance—so the blind spot belongs to *specialised* critics, and is caught by any perceptually-capable scorer (prompted VLM, VL reward, or CLIP quality read).

It abstains instead of hallucinating geometry. Under image-only inspection, the finetuned model stays at 0.50 on G2–G6 with *zero* false positives: trained with image-only abstention targets, it declines to hallucinate geometry from pixels and leaves those classes to the linter—the designed behavior, and the reason the hybrid's geometry column comes from the linter rather than the VLM.

Honest negatives. Two results cut against an over-broad reading. First, *deck-scope* semantics fail pointwise: the finetuned model is degenerate on S2 (recall 1.0, specificity 0.0—always flags) because the SFT mix contained no clean-*deck* negatives, and it abstains on the held-out S5; deck-scope classes were never given the pairwise head the diagnosis prescribes for this failure mode, and fixing this needs a retrain we did not run. Second, there is a *sim2real gap*: on image-only SlideAudit, the finetuned model's synthetic S4 strength (0.97) does not transfer (0.50); only the zero-shot 30B shows any real image-only density signal (recall 0.53 vs. the finetuned 0.075, $p = 1.1 \times 10^{-5}$). Scale, not finetuning, carries bare-pixel transfer—and the finetuned model learned a defect *distribution*, not a universal notion of slide quality. We make this concrete next.

7 External validity

A real pre-sales agent deck. We run the same five-level examiner gradient on a real 16-slide proposal produced by a deployed PowerPoint agent. Its verifiable defect is 20 unfilled template placeholders ("add a title here") across 7 of 16 slides—a semantic-completeness defect the geometry linter is blind to by construction (the boxes are geometrically fine). The result reinforces the thesis from an unexpected angle: the zero-shot 30B is the best detector (recall 1.0, names the placeholder on 4/7 slides, scores them 0.03 vs. 0.64 for clean), while the *finetuned* 8B—top of the intrinsic-quality ranking—is among the *worst* on this out-of-distribution defect (names it 0/7), and the linter's apparent recall is spurious geometry noise with zero placeholder mentions. The ranking on a novel defect is by general-VLM ability, not by the specialist's in-distribution quality: the examiner learned a distribution, not universal quality. Feeding the flagged

Table 4: Page-semantic inspection, balanced accuracy [95% Wilson] on paired clean (best modality). The finetuned 8B examiner beats both zero-shot baselines, including the 30B, *in-distribution*. Deck-scope S2/S5 are an honest negative (text). n = paired positives (2-AFC rows = forced-choice pairs); the S6 2-AFC rests on only 12 pairs (wide interval).

class	**ft-8B**	zs-8B	zs-30B	n
S1 title/body	**1.000** [.95,1]	0.778 [.69,.85]	0.933 [.87,.95]	36
S4 density	**1.000** [.97,1]	0.529 [.50,.58]	0.774 [.69,.84]	60
G1 overflow (2-AFC)	**1.000** [.97,1]	0.975 [.93,.99]	0.700 [.61,.77]	120
S6 image/text (2-AFC)	**1.000** [.76,1]	1.000 [.76,1]	0.500 [.25,.75]	12
S2 narrative (deck)	0.500† [.45,.53]	0.646 [.53,.72]	0.549 [.42,.66]	36
S5 missing-logic (deck)	0.500 [.47,.53]	0.500 [.47,.53]	0.567 [.45,.68]	60

† degenerate (always-flag): the SFT mix had no clean-*deck* negatives. See text.

placeholders back to the agent's own generator for one revision pass drives the verifiable defect from 20 to 0.

The structured attribution holds on real layouts. The body runs the A/B/C attribution on synthetic slides; the cleanest discrimination needs it on *real* geometry with a *real* oracle, which seemed to require human or model annotation. We obtain it without either: Zenodo10K [18] ships real CC-licensed `.pptx` with intact element XML, so `python-pptx` extracts a geometry/text/style oracle *lossless with respect to the declared IR* straight from the source file (the IR→render gap that G7 exploits is exactly what this oracle does *not* capture, by design)—no human, no self-annotation bias. From 26 real decks we inject one single-shape defect per slide *in PPTX XML space* (overflow, overlap, alignment offset, font, margin) and render both the clean and the defective one-slide deck with the same real renderer (LibreOffice), so the paired pixel diff isolates the injected defect on real pixels (209 paired slides, 5 classes, 3 VLMs / 3 families). Two results follow. First, the §4 "snap absorbs 45%" hazard is *template-specific*: real free-form decks absorb only **7%** (render-fidelity 0.93), so the perturbation is faithfully present—a positive control on that claim. Second, *all three* attribution outcomes appear on real geometry: coarse defects (overflow, overlap) are **image-sufficient** (balanced accuracy A 0.70–0.74, structure adds nothing); a margin bleed is a **perception bottleneck the oracle rescues** for the weaker VLMs (A 0.59 → B 0.70)—the exact "missing structure, not missing capability" cell; and fine alignment stays near chance in *every* channel for *every* model (A = B = C ≈ 0.50; the internal-contrast re-run at a supra-threshold 44 px offset reproduces this, A 0.54/B 0.51/C 0.52, n=41). This is *not* a hard capability floor—on clean synthetic slides the same alignment defect recovers under targeted/relative elicitation once it clears acuity (Fig. 3)—but on real cluttered layouts the misaligned element cannot be isolated among many genuine ones and specificity collapses, so for declared geometry the symbolic linter (which reads the coordinates) remains the reliable detector regardless (Fig. 9).

Open-world: structure recovered from pixels does not restore the linter. The hybrid's symbolic engine needs the declared IR a `.pptx` carries; a third-party deck exported to PDF/PNG ships none, so the natural question is whether *recovering* element structure from the rendered pixels can re-arm the linter. We test it directly: a transformers-native document-layout detector (PP-DocLayoutV2) returns region boxes on each render, class-agnostic NMS removes the detector's cross-class duplicate boxes (which would otherwise manufacture phantom overlaps), and the boxes—normalised into the IR coordinate frame—feed the *same* geometry linter at its shipped operating point. On the real-layout set (GT IR *and* render available, so the full ladder and a recovery-fidelity IoU coexist) the answer is a clean, pre-registered negative (Fig. 10): recovered structure leaves the linter at chance on G1/G3/G4 (overflow needs text capacity, alignment needs declared positions, font needs point sizes—none survive a pixel→box projection), and rescues only overlap partially (G2 balanced accuracy 0.54 vs. native-IR 0.67 vs. VLM-only 0.87). The detector *over-segments*—≈ 2× more boxes than the GT IR, recovery recall at IoU≥0.5 only 0.30–0.39—so genuine element collisions are split apart and overlap recall falls 0.78 → 0.29. The same picture holds on image-only SlideAudit (no IR at all): the recovered linter clears chance only on G2 (0.55) while the C3-elicited VLM reaches 0.94. Two points sharpen the deployment scope. First, on real third-party decks even *native* IR is weakened—the decks carry no expected-position/font bookkeeping, so the linter's G3/G4 rules are also at chance, and its margin/overlap rules *over-fire* on tight real layouts (specificity 0.11/0.56), eroding the near-zero-false-positive advantage it enjoys on clean synthetic IR. Second, as a direct consequence the VLM-only engine *dominates every coarse class* on real decks. The linter's edge is an in-distribution, clean-native-IR phenomenon; open-world—recovered or not—the deployable critic is VLM-bound. This bounds the contribution without touching the diagnostic thesis: the full symbolic–neural critic is for IR-owning

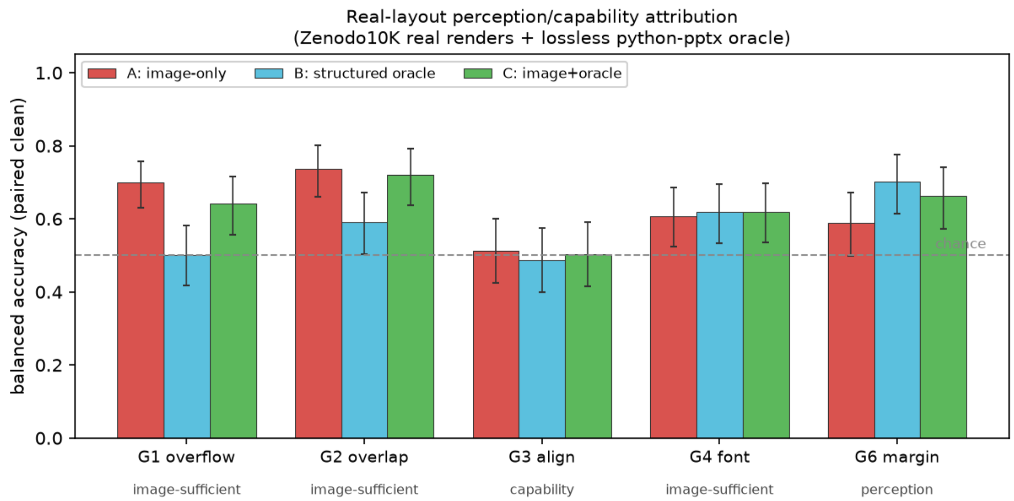

Figure 9: The perception/capability split reproduces on real layouts (ZENODO10K real renders + lossless `python-pptx` oracle; mean over 3 VLMs / 3 families, balanced accuracy on paired clean; error bars = 95% Wilson on counts pooled across models, *n*=76–90 paired per class). Coarse defects (G1 overflow, G2 overlap) are image-sufficient; G6 margin is a *perception* bottleneck the structured oracle rescues (B>A); G3 alignment stays near chance in every channel on these real cluttered decks (A=B=C≈0.5)—not a hard capability floor (it recovers on clean slides above threshold, Fig. 3) but unreliable enough on real layouts that it must route to the linter. Intervals on G3/G4/G6 include chance, consistent with their small per-class *n* (see released reports).

generation agents, and for bare third-party pixels the critic is the C3-elicited VLM.

Does a better examiner help downstream? A bounded effect, and why (E6). We also asked whether examiner quality transfers to generation: vary the examiner across five intrinsic-quality levels, feed its critique into a self-refine loop, and measure an independent model-free quality. The direction is the hypothesized one—examiner quality correlates +0.66 with refinement gain, ranking the linter last and the strong semantic examiners first—but the magnitude is sub-1% because a mainstream generator floors most briefs at the first draft, leaving little headroom. A natural rebuttal is that the effect would be material with headroom, so we re-ran the *identical* gradient with a weak local generator (Qwen3-VL-4B). The weak generator is genuinely unfloored—1.43× the headroom of the strong one—yet the effect does not grow, it *vanishes*: the examiner-quality↔gain correlation collapses from +0.66 to 0.00 and *no* examiner buys any gain (Fig. 11A). The reason is *where* the headroom sits: it is coverage-dominated (deck completeness; component 0.19), and coverage is critiqued by neither the geometry linter nor the injected-defect examiner—a perception/defect critic has nothing to say on that axis. A controlled actionability A/B (generator, task, and seed fixed; only the critique varied) confirms the null is an *axis mismatch*, not a revision incapacity: the real geometry-clean critique moves quality by +0.00, while an explicit coverage critique moves it by +0.32 (coverage +0.81), the same weak model adding the named missing sections (Fig. 11B). The signal is therefore squeezed from both sides—a strong generator leaves no headroom, a weak one leaves headroom off the critique axis—so the verifiable evidence for the examiner is its intrinsic (Secs. 5.1, 6) and case-study performance, not this loop. A second optimization-efficiency vehicle reverses sign as an artifact of its metric, consistent with this reading.

8 Limitations

We *do* run the structured (image+oracle) evaluation on real slides that separates "missing structure" from "missing capability" on real geometry (§7): a lossless `python-pptx` oracle over real CC-licensed decks sidesteps both the annotation labour and the self-annotation bias, and the split reproduces—fine alignment stays near chance in every channel on real cluttered decks (it recovers on clean slides above a magnitude threshold, but real clutter defeats it) while a margin bleed is structure-rescuable. Its scope is honest: the defects are *injected* (real geometry, real renderer, real oracle, but not naturally-occurring), so the third-party SLIDEAUDIT set (image-only, naturally-defective) remains our complementary real probe. We audit five published reward scorers, two per audited category (general-multimodal,

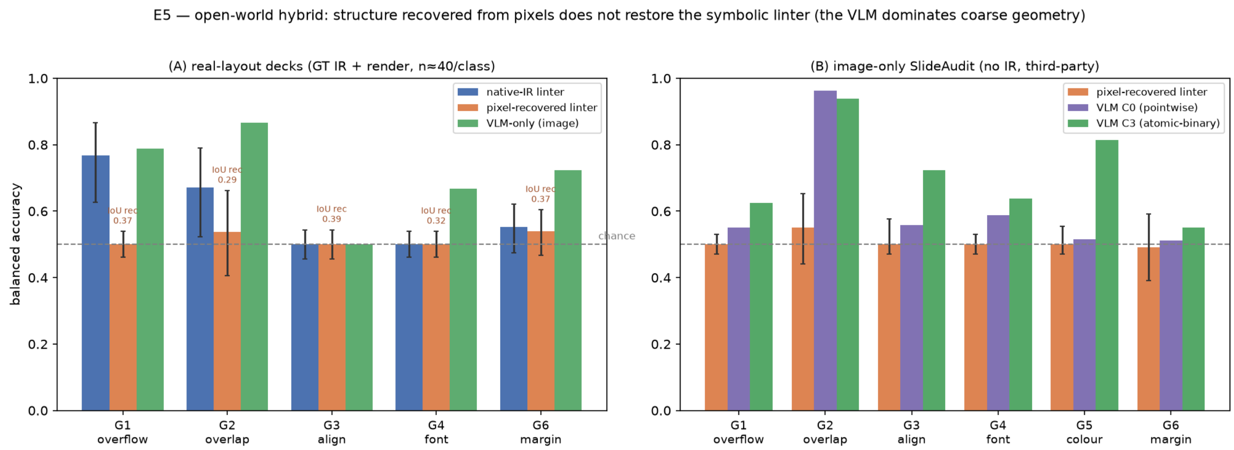

Figure 10: Open-world hybrid (E5): structure recovered from pixels does not restore the symbolic linter. A document-layout detector (PP-DocLayoutV2) recovers element boxes from the render; the *same* geometry linter runs on them. **(A)** real-layout decks (GT IR + render, n≈40/class): native-IR linter vs. pixel-recovered linter vs. VLM-only floor, Wilson CIs; recovered structure rescues only overlap (G2) and is silent on fine geometry, while the VLM dominates every class. **(B)** image-only SlideAudit (no IR, third-party): the recovered linter clears chance only on G2; the C3-elicited VLM is far stronger. The linter's near-zero-FP advantage requires clean native IR.

aesthetic), with the document category at one (a second document/design reward, PosterReward, needs an ms-swift `seq_cls` runtime we did not stand up, as does an InternLM-family reward whose PLoRA image path failed to load); the category-level reads are therefore solid for general-multimodal and aesthetic and instance-level for document. Our attribution protocol shares its core with concurrent perception/reasoning work, differentiated by domain, the lossless oracle, and the loop to a critic. And the downstream effect is bounded on both sides—small under a strong generator (no headroom) and absent under a weak one (its headroom is coverage, off the examiner's perception/layout axis; §7, Fig. 11). None of these is hidden; each is stated where it bites.

Two further caveats temper the central claim and we state them plainly. First, the forced-choice and atomic-binary elicitations supply a contrastive reference and a named target that the pointwise format withholds, so one must ask whether the 0.50 → 1.00 recovery is *elicitation* or *reduced task difficulty*. We ran the clean ablation rather than leaving it to future work (the single-slide, defect-type-named *absolute* condition C0_named, plus a clean-vs-clean guess-floor control; Sec. 5.1, Fig. 6), and the answer is defect-dependent: for the constructed render class G7 naming alone recovers detection (format suppression), while for declared G1/S6 the recovery genuinely requires the paired reference (availability-of-reference). We therefore make the strong mechanism claim only for G7 and its C3 mirror, and label the G1/S6 forced-choice effect as reference-availability; the guess-floor control (capable models answer "tie" on 92–100% of clean pairs) rules out a forced-pick artifact for both. Second, the 8/9 coverage (0.89 for linter+VLM-C3; 0.88 for the pre-registered routed hybrid) is a *native-structure* number; on image-only real decks (no IR for the linter) the hybrid degrades to its VLM-only mode, so that figure is an upper bound for IR-owning generation agents, not an open-world guarantee. We do not leave this to assertion: recovering element structure from the pixels with a layout detector and re-running the linter does *not* close the gap (§7, Fig. 10)—it rescues only overlap, partially, and the VLM-only engine dominates every coarse class on real decks, so the open-world critic is VLM-bound.

9 Conclusion

A slide-defect critic should not be one model. Telling apart the two blindnesses—a magnitude-gated acuity floor versus format-suppressed perception—turns a vague "VLMs are bad at layout" into an actionable routing rule: rules to the linter and perceivable-but-suppressed classes to a VLM under an elicitation that the diagnosis proves recovers them. The E3 control makes the rule simpler: linter plus one C3-prompted VLM covers 8/9 classes, matching the pre-registered routed hybrid and beating either single engine. A falsifiable render class G7, audited over five published rewards (two per category), shows that the linter's blind spot is shared by a document reward and an artistic-aesthetic head but *not* by either of two general-multimodal rewards or a perceptual-quality probe—so G7-detection tracks perceptual

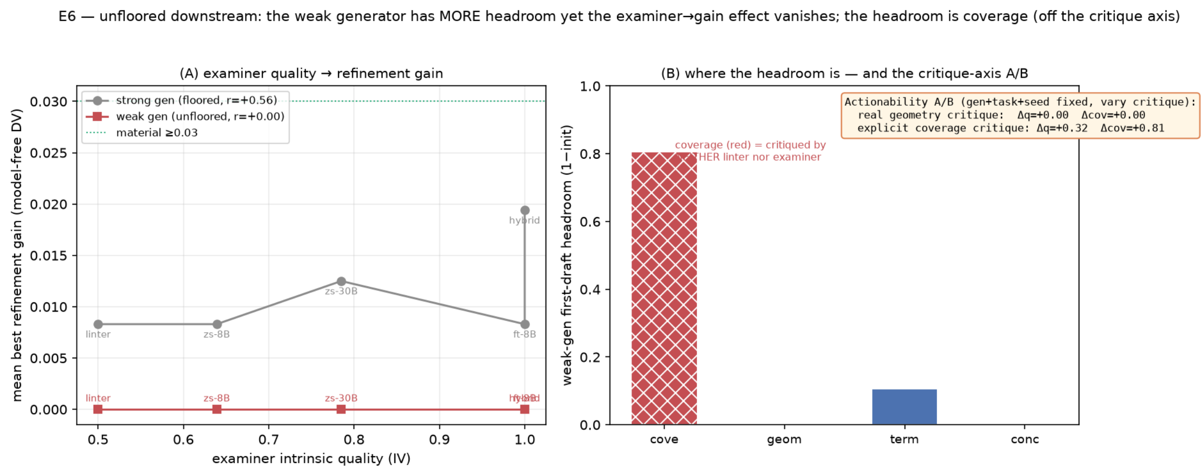

Figure 11: Unfloored downstream (E6): more headroom, no gain—a critique-axis mismatch. (A) examiner intrinsic quality vs. best refinement gain on a model-free quality. The strong (flooring) generator gives the hypothesised slope (r=+0.56) at tiny magnitude; the weak generator—despite 1.43× the first-draft headroom—is flat (r=0.00), no condition reaching the material threshold. **(B)** the weak generator's headroom by dimension is dominated by *coverage* (red), which neither the linter nor the examiner critiques; the inset actionability A/B (same generator/task/seed, only the critique varies) shows the real geometry critique recovers nothing (Δq=+0.00) while an explicit coverage critique recovers Δq=+0.32 (Δcov=+0.81)—so the null is an axis mismatch, not an inability to revise.

capability, not the training-domain label, and G7 needs a perceptually-capable engine, prompted or reward-headed, not a specialised one. The recovery replicates across four of the six VLMs tested and four families and transfers to real slides, which makes the central claim a property of the *elicitation* × *defect* interaction rather than of any one model. The unit of contribution is the diagnosis-to-routing pipeline and the falsifiable class, not a leaderboard number—and the negatives we kept in are part of the evidence.

Reproducibility

All analysis scripts, the defect injector, the two linters, the elicitation harness, the hybrid router, the multi-reward audit adapters, the real-layout (Zenodo10K) IR/inject/render pipeline, and the per-Part reports are released. Experiments ran on 4×RTX 3080 (20 GB) with vLLM-served open VLMs (Qwen3.5/3.6, Gemma4, InternVL3.5, Ovis2.5), a QLoRA-finetuned Qwen3-VL-8B, and five published reward scorers (DocReward-3B; Skywork-VL-Reward-7B and PickScore; a LAION aesthetic predictor and CLIP-IQA) behind a uniform adapter; serving recipes (tensor-parallel, AWQ, disabled thinking, fp8-KV caveats on Ampere) are in the repository. Weights, renders, and raw rollouts are not redistributed. Every balanced-accuracy cell in the tables and figures carries its 95% Wilson interval and per-cell n; the full family of paired tests is reported with Holm (FWER) and Benjamini–Hochberg (FDR) multiplicity correction over all 61 reported significance tests (released multiplicity report), and headline claims are stated as surviving or honestly downgraded under that correction.

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